

Cramér–Rao analysis of orientation estimation: viewing geometry influences on the information conveyed by target features

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A methodology for analyzing an imaging sensor's ability to assess target properties is developed. By the application of a Cramér–Rao covariance analysis to a statistical model relating the sensor measurements to the target, a lower bound can be calculated on the accuracy with which any unbiased algorithm can form estimates of target properties. Such calculations are important in understanding how a sensor's design influences its performance for a given assessment task and in performing feasibility studies or system architecture design studies between sensor designs and sensing modalities. A novel numerical model relating a sensor's measurements to a target's three-dimensional geometry is developed in order to overcome difficulties in accurately performing the required numerical computations. The accuracy of the computations is verified against simple test cases that can be solved in closed form. Examples are presented in which the approach is used to investigate the influence of viewing perspective on orientation accuracy limits. These examples are also used to examine the potential accuracy improvement that could be gained by fusing multiperspective data. © 2003 Optical Society of America

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1. INTRODUCTION

One of the most common uses of imagery is the assessment of the properties of the imaged target and its sub-components. Properties that are directly related to an image include size, shape, position, orientation, temperature, and reflectance characteristics, to name a few. In the process of designing a sensor or planning a measurement operation, it is important to understand and predict how a sensor's characteristics and the method of its employment influence the accuracy with which the desired target assessments can be made. It is also of interest to understand the performance benefit that might be achieved by fusing data from multiple sensing sources. Examples of data fusion include combining information from images taken at multiple perspectives and times and/or from a mix of sensing modalities such as conventional passive optical, hyperspectral, ranging laser radar, radar, and range-Doppler imaging measurements.

Assessment accuracies are often characterized by analyzing the noise properties of specific estimation algorithms or by testing algorithms against simulated or experimental data and comparing the results against the known or otherwise inferred state of the target.^{1–8} This approach is most useful in predicting the accuracy of a specific target, sensor, and processing algorithm combination. The accuracies obtained will, however, be colored by the optimality of the algorithm. To use this approach in determining how sensor and target characteristics limit assessment accuracy, the algorithms employed must have optimality characteristics such as those exhibited by maximum-likelihood, maximum *a posteriori*, and minimum-mean-square-error (MMSE) estimators. Simulations using optimal estimators have been widely and suc-

cessfully used to characterize target assessment accuracies on tasks such as detection, recognition, location, and orientation estimation.^{1,2,9–17} The development and the implementation of such algorithms can be difficult and time-consuming and may involve significant computational requirements, especially if many repetitions are needed to build up adequate performance statistics. This approach is convenient, however, if an end goal is to realize an optimal estimation algorithm.

This paper addresses the discussed analysis needs by using the approach of the Cramér–Rao lower bound (CRLB).^{18,19} Because this accuracy bound is derived from a probabilistic and presumably physics-based model of the target sensor interaction and measurement noise processes, it represents a fundamental limit that is independent of the ability or the requirement to implement an optimal estimation algorithm. The CRLB simply gives the lowest mean square error (MSE) achievable by the theoretically optimal unbiased estimator. This is powerful for performing feasibility studies and system architecture design studies, since it permits sensor performance to be analyzed without knowledge of an optimal estimation algorithm. The CRLB is also a useful benchmark for comparing actual algorithms against, since it describes a limit on the best possible performance. It must be kept in mind, though, that if in practice an efficient estimation algorithm is not implemented, performance could be significantly worse than the limit indicated by the CRLB. This is strictly true only if the actual physics of the target/sensor system truly follows the forward model used to compute the CRLB. Differences could permit the real system to perform better than the CRLB. Typically, however, increasing the fidelity of the model to

include more real-life uncertainties in the system drives the CRLB up. Since it is virtually impossible to include all actual uncertainties in the model, the CRLB is generally optimistic. In many cases, an estimator that achieves the CRLB (an efficient estimator) does not exist. Other bounds may be derived—for example, the Bhattacharyya, Bobrovsky–Zakai, and Weiss–Weinstein bounds—that are equal to or greater than the CRLB and thus may provide tighter limits on the accuracy that the optimal estimator could achieve.^{18–20}

An advantageous feature of CRLBs is that they are solely a function of the local derivatives of the Bayesian probability density function (pdf) used to describe the measurement process as averaged over the noise processes. A significant body of work has been performed with the use of bounds on the MSE of various error metrics that have many of the same advantages as those of the CRLBs for evaluating theoretical limits on target assessment accuracy.^{21–24} MSE bounds typically involve evaluating integrals of the error metric over the space of possible parameter values and weighted by the posterior pdf. The CRLB provides an alternative approach, which in many cases will require fewer computational operations because of the fact that it is solely a function of local derivatives. This is especially important in addressing estimation problems that involve many degrees of freedom. Though similar, Cramér–Rao and MSE bounds offer slightly different and complementary interpretations of estimation accuracy limits.^{18–20} Relationships between CRLBs and MSE metrics on target assessment under asymptotic limits such as high signal-to-noise ratio have been examined by Grenander *et al.*²³ and Cooper and Miller.²⁴ It should be pointed out that the application of CRLBs is restricted to parameter spaces that are flat or limiting situations in which nonflat parameter spaces such as orientation Euler angles can be approximated as flat. MSE-type bounds more naturally accommodate such issues.^{21–23,25} This issue is discussed in further detail in Subsection 2.B.

Because the pdfs from which Cramér–Rao and MSE bounds are derived are often easily generalized²⁶ to express the joint statistics of measurement sequences and multiple sensors, they naturally address analysis of the potential performance benefits of data fusion. This permits the performance limits on assessing the same target features using fundamentally different measurement modalities to be compared on a common footing. It also provides a method for understanding the ways in which combinations of multiple measurements may provide complementary information. This use of CRLBs is demonstrated in Section 6, in which the potential accuracy gains of fusing multiperspective data for target orientation estimation is analyzed. Numerous other examples of this feature of probability-model-based approaches exist in the literature both as a method of accuracy prediction and as a framework for development of data fusion algorithms.^{1,2,11,12,15,16,22} Through the use of Fisher information, which is a function of the Bayesian model and from which the CRLB is derived, this paper also examines how information regarding target orientation is exhibited within an image. Insights on how the location of features on a target influences the information provided re-

garding different orientational aspects are developed through illustrative examples presented in Sections 4 and 6 by using simple rectangular targets and complex three-dimensional (3-D) targets, respectively.

The starting point for CRLB calculations is a forward probabilistic model describing the statistical relation of the imaging measurement to a state vector of target properties. For common additive noise sources with Gaussian or Poisson statistics, the required computations primarily involve calculating derivatives of the mean image measurement with respect to the object state parameter values. Because of the highly nonlinear relations between the two-dimensional (2-D) image and 3-D target structure, accurate numerical calculation of the derivatives is challenging. However, once an adequate target/sensor model is developed, it is straightforward to analyze assessment bounds for any combination of target and sensor properties and for multiple images and sensor types. The calculated bound corresponds to the accuracy with which the estimated target properties will tend to match the true stipulated values. This implies that the bound is specific to the actual values of all target properties, to the imaging geometry (e.g., range, illumination, and perspective), and to the sensor characteristics [e.g., noise sources, aperture size, focal length, instantaneous field of view (IFOV), spectral sensitivity, and exposure time]. IFOV refers to the angular portion of the field of view (FOV) subtended by one focal-plane-array (FPA) element, that is, the actual width of the element divided by the focal length of the system.

In exploring the use of CRLBs for assessment accuracy characterization, we focus on estimating properties of an incoherently illuminated target against a uniform background using gray-scale imagery as would be produced by a conventional passive electro-optic imaging system. This encompasses a wide class of problems including assessing properties of bodies in space, or in earth's atmosphere against a uniform background, and for many earth-bound situations such as surveillance or robotic vision (so long as clutter is not significant). The dominant noise sources are assumed to be a combination of photon noise from target, background, and ambient light sources, along with FPA readout noise. General CRLB expressions are derived in Section 2 for an arbitrary set of target and sensor parameters. Accurate treatment of more complicated noise sources such as clutter, coherent-illumination-related speckle, and turbulence-induced speckle would require further generalization of the imaging model presented here.

The illustrative estimation problem focused on in this paper is target orientation. Section 3 discusses the difficulties of calculating partial derivatives on the nonlinear relationships of image plane measurements to orientation of targets with complex 3-D shapes and lighting geometries. It then presents a novel approach to overcoming this problem by development of a highly accurate rendering model that generates smooth and continuous changes in the measurement values with respect to small changes in target shape and orientation. The approach is based on an analytic method of calculating the radiant exitance from discrete surface elements on the target's surface and incident on each pixel of the sensor's FPA. The accuracy

of the approach is verified in Section 4 by comparing numerical results against an exact closed-form expression derived for a simple target. Sensitivity of the calculations to the step size used in the finite-difference approximations is also examined.

Section 5 explores the impact of nuisance parameters. These correspond to unknown information about the target or the sensor that is not really of concern except for the impact of the added uncertainties on the ability to estimate the quantities of principal interest. However, inclusion of all the relevant target and system uncertainties is quite important to the CRLB analysis approach.^{10,14,19,21,22,24,27,28} If there are cross correlations in the measurement between the parameters of principal interest and other quantities not included in the analysis, the calculated bound will be artificially low and thus decreased in relevance to the real-world problem.

Section 6 demonstrates the presented methodology by analyzing the decrease in the CRLB on the accuracy with which a satellite's orientation can be estimated when several images taken from different perspectives are considered jointly. This section also examines the influence of viewing perspective on bounds for estimating different aspects of a target's orientation.

2. MATHEMATICAL DEVELOPMENT

A. Imaging Sensor Cramér–Rao Lower Bound

Assume that the vector $\xi = \{\xi_1, \xi_2, \dots, \xi_k, \dots, \xi_N\}$ specifies all relevant target/sensor system parameters that are unknown. The Cramér–Rao bound describes a lower limit on the accuracy achievable (minimum error variance) in forming an unbiased estimate of ξ_k from a measurement vector d ^{18,19}:

$$\langle (\hat{\xi}_k - \xi_k)^2 \rangle \geq \text{CRLB}\{\hat{\xi}_k|\xi\} = [F_D^{-1}(\xi)^{-1}]_{kk}, \quad (1)$$

$$\begin{aligned} [F_D(\xi)]_{kl} &= -\left\langle \frac{\partial^2 \ln p(d|\xi)}{\partial \xi_k \partial \xi_l} \right\rangle \\ &= \left\langle \frac{\partial \ln p(d|\xi)}{\partial \xi_k} \frac{\partial \ln p(d|\xi)}{\partial \xi_l} \right\rangle. \end{aligned} \quad (2)$$

In these equations, $\hat{\xi}$ is the estimated value of ξ , $F_D(\xi)$ is the Fisher information matrix representing information obtainable from the data, and the averages are performed over the noise statistics. The quantity $p(d|\xi)$ is the pdf for obtaining measurement d given values of ξ corresponding to a particular system state. This function must be specified such that it accurately models the system of interest.

Typically, in addition to the principal parameters that one wishes to estimate, the vector of unknowns ξ includes additional uncertainties in the system. To estimate the principal parameters, either we must specify values of these additional “nuisance” parameters by using whatever *a priori* information is available and integrate the nuisance parameters out of the pdf (possibly weighted by any *a priori* statistics) or we must jointly estimate the full parameter set. These additional uncertainties can have a significant influence on accuracy bounds. To obtain realistic results, it is important for their influence to be con-

sidered in the analysis along with the influence of any *a priori* information that can be employed. The influence of *a priori* knowledge is easily incorporated into the bound by adding a second term $F_P(\xi)$ to relation (1), which characterizes the additional information provided by the prior $p(\xi)$:

$$[F_P(\xi)]_{kl} = -\left\langle \frac{\partial^2 \ln p(\xi)}{\partial \xi_k \partial \xi_l} \right\rangle = \left\langle \frac{\partial \ln p(\xi)}{\partial \xi_k} \frac{\partial \ln p(\xi)}{\partial \xi_l} \right\rangle. \quad (3)$$

The modified bound is²⁹

$$\langle (\hat{\xi}_k - \xi_k)^2 \rangle \geq [\{F_D(\xi) + F_P(\xi)\}^{-1}]_{kk}. \quad (4)$$

When the *a priori* statistics of ξ are jointly Gaussian with covariance matrix Λ_ξ and mean $\langle \xi \rangle$, i.e.,

$$\begin{aligned} p(\xi) &= [(2\pi)^{N/2} |\Lambda_\xi|^{1/2}]^{-1} \\ &\times \exp[-\frac{1}{2}(\xi - \langle \xi \rangle)^T \Lambda_\xi^{-1} (\xi - \langle \xi \rangle)], \end{aligned} \quad (5)$$

then the *a priori* information is simply described by the inverse of the covariance matrix of ξ , i.e., $F_P = \Lambda_\xi^{-1}$. The influence of nuisance parameters and *a priori* information on target parameter estimation accuracy bounds will be explored further in Section 5.

1. Noise Statistics

To develop the imaging CRLB further, we must now specify the noise statistics described by $p(d|\xi)$. We consider imaging systems for which the measurement noise is statistically independent between each FPA element and is an additive sum of Poisson and Gaussian noise sources:

$$d(\bar{x}) = \text{Poisson}\{g(\bar{x}, \xi)\} + n_{\sigma_{\bar{x}}}. \quad (6)$$

In Eq. (6), $n_{\sigma_{\bar{x}}}$ is a zero-mean Gaussian random variable with standard deviation $\sigma_{\bar{x}}$, Poisson $\{\Xi\}$ denotes a Poisson random variable with mean Ξ , and $\bar{x} = (x, y)$ indexes the grid of elements on the imaging system's FPA. The function $g(\bar{x}|\xi)$ specifies the mean number of photoelectrons plus dark-current electrons detected at sensor element $\bar{x}$ of the image measurement when the target/sensor system state is ξ , and thus it embodies the relation of the ensemble-average image to all target/sensor parameters.

Note that both the Poisson and Gaussian terms may each include contributions from several signal and noise processes that are in addition to the target, such as target background, stray light, dark current, and electronics noise. Noise associated with digital quantization of the signal has been neglected. This approximation is reasonable as long as the quantization step size is significantly smaller than either the Poisson or Gaussian noise components and the signal does not saturate. The described noise model is appropriate for incoherent image measurements using a wide class of sensors such as CCD, complementary metal-oxide semiconductor, and photomultiplier arrays along with many infrared sensors.^{14,15,30,31} Accurate treatment of noise sources with pixel-to-pixel correlations, such as laser speckle, turbulence-induced scintillation, and backgrounds with random structure (such as clutter), would require further generalization of the imaging model presented here.

2. Cramér–Rao Lower Bound for Poisson Noise

Examining first the simplified case for which the measurement vector d includes only a Poisson noise contribution (i.e., $\sigma_{\bar{x}}^2 = 0$), we obtain the following conditional probability pdf:

$$p(d|\xi) = \prod_{\bar{x} \in \mathcal{F}} \frac{\exp[-g(\bar{x}|\xi)] g(\bar{x}|\xi)^{d(\bar{x})}}{d(\bar{x})!}. \quad (7)$$

In Eq. (7), $\mathcal{F}$ denotes the set of FPA elements in the FOV. If we substitute Eq. (7) into Eq. (2) and employ the fact that for Poisson statistics the variance of $d(\bar{x}|\xi)$ is equal to its mean $g(\bar{x}|\xi)$, it is straightforward to show that

$$[F_D(\xi)]_{kl} = \sum_{\bar{x} \in \mathcal{F}} \frac{\frac{\partial g(\bar{x}|\xi)}{\partial \xi_k} \frac{\partial g(\bar{x}|\xi)}{\partial \xi_l}}{g(\bar{x}|\xi)}. \quad (8)$$

3. Cramér–Rao Lower Bound for Gaussian Noise

If instead we assume that the noise associated with the measurements at each FPA element $\bar{x}$ consists solely of a Gaussian component with variance $\sigma_{\bar{x}}^2$, i.e.,

$$d(\bar{x}) = g(\bar{x}, \xi) + n_{\bar{x}}, \quad (9)$$

then the measurement d is described by the following conditional pdf:

$$p(d|\xi) = \prod_{\bar{x} \in \mathcal{F}} \frac{1}{\sqrt{2\pi\sigma_{\bar{x}}^2}} \exp[-|d(\bar{x}) - g(\bar{x}|\xi)|^2/(2\sigma_{\bar{x}}^2)]. \quad (10)$$

Upon substituting Eq. (10) into Eq. (2), we find that

$$[F_D(\xi)]_{kl} = \sum_{\bar{x} \in \mathcal{F}} \frac{\frac{\partial g(\bar{x}|\xi)}{\partial \xi_k} \frac{\partial g(\bar{x}|\xi)}{\partial \xi_l}}{\sigma_{\bar{x}}^2}. \quad (11)$$

When $k = l$, each term of the sums in Eqs. (8) and (11) expresses the Fisher information gained from an individual FPA element $\bar{x}$ about a single state parameter ξ_k . From this, we see that the Fisher information about parameter ξ_k is proportional to the sensitivity of a pixel measurement to variations in that state parameter and inversely proportional to the measurement noise. Because of statistical independence of the noise at each pixel, the total Fisher information is simply the sum over the Fisher information obtained from all detector elements. When $k \neq l$, each term in Eq. (8) or (11) expresses the joint Fisher information for system parameters ξ_k and ξ_l contributed by each FPA element.

4. Cramér–Rao Lower Bound for Additive Combination of Poisson and Gaussian Noise

When the noise is an additive combination of Poisson and Gaussian sources, the pdf becomes more complicated. However, it is often a reasonable treatment of this situation³² to use the following approximate noise model:

$$d(\bar{x}) = \text{Poisson}\{g(\bar{x}, \xi) + \sigma_{\bar{x}}^2\} - \sigma_{\bar{x}}^2, \quad (12)$$

which matches the mean and the variance of the combined noise. Within this approximation,

$$[F_D(\xi)]_{kl} = \sum_{\bar{x} \in \mathcal{F}} \frac{\frac{\partial g(\bar{x}|\xi)}{\partial \xi_k} \frac{\partial g(\bar{x}|\xi)}{\partial \xi_l}}{g(\bar{x}|\xi) + \sigma_{\bar{x}}^2}. \quad (13)$$

Note that the denominator of the fraction in Eq. (13) is the noise variance of pixel $\bar{x}$.

5. Cramér–Rao Lower Bound for Multiple Images

Let us now generalize the result to the case for which d corresponds to multiple images, taken either as a temporal sequence from one sensor and/or by multiple sensors that may each employ different imaging modalities. Assuming that the noise for each sensor is statistically independent and using the described method of approximating the Poisson and Gaussian noise combination by a biased Poisson process, we obtain the following pdf:

$$\begin{aligned} p(d + \bar{\sigma}^2|\xi) &= \prod_q \prod_{\bar{x} \in \mathcal{F}_q} \\ &\times \frac{\exp[-g_q(\bar{x}|\xi) + \sigma_{\bar{x},q}^2] [g_q(\bar{x}|\xi) + \sigma_{\bar{x},q}^2]^{d_q(\bar{x}) + \sigma_{\bar{x},q}^2}}{[d_q(\bar{x}) + \sigma_{\bar{x},q}^2]!}, \end{aligned} \quad (14)$$

where q is used to index the image set and $\mathcal{F}_q$ represents the FOV for image q . This final generalization results in

$$[F_D(\xi)]_{kl} \cong \sum_q \sum_{\bar{x} \in \mathcal{F}_q} \frac{\frac{\partial g_q(\bar{x}|\xi)}{\partial \xi_k} \frac{\partial g_q(\bar{x}|\xi)}{\partial \xi_l}}{g_q(\bar{x}|\xi) + \sigma_{\bar{x},q}^2}, \quad (15)$$

which is just the sum of the Fisher information matrices corresponding to each image. Because of this simple result, it is easy to calculate the potential performance benefits that could be achieved by fusing data from multiple image measurements and imaging modalities. It can be shown that combining information from multiple independent measurements always reduces the CRLB on every element of the vector parameter ξ below that obtained from any single measurement.²⁷ This is a direct consequence of the data processing theorem for Fisher information.³³

It is of interest to note that when the Gaussian noise component can be neglected, such as in the photon noise limit (i.e., $\sigma^2 \ll g \forall \{\bar{x} \in \mathcal{F}_q; q = 1, 2, \dots\}$), the Fisher information scales linearly with the light level of the image. Correspondingly, estimation accuracy scales as the square root of the inverse of the light level. In the opposite limit, for which background noise dominates ($\sigma^2 \gg g \forall \{\bar{x} \in \mathcal{F}_q; q = 1, 2, \dots\}$), the Fisher information scales in proportion to the square of the light level, and estimation accuracy scales as its inverse.

Later sections of this paper will focus on sensors that measure the intensity distribution of light reflected off a target as a function of angle about the line of sight (LOS). Photon- and sensor-noise-dominated gray-scale images taken by a conventional passive electro-optic system operating at visible wavelengths are common examples that follow this model. Equation (14) and relation (15) also

apply to IR and multispectral imaging modalities and may be appropriate to other sensor types as well, including nonimaging modalities if the index $\bar{x}$ of the measurement parameter $d(\bar{x})$ is constructed as a vector instead of as a 2-D array.

B. Interpretation Issues of Cramér–Rao Lower Bound for Orientation Estimation

The target assessment task focused on in this paper is target orientation. The classical Cramér–Rao bound formulation applies only to flat Euclidean spaces in which each element of the vector parameter may take on any real value. Alternatives have been developed that provide a rigorous treatment of accuracy bounds on parameter spaces with curved geometries.^{12,13,22–25} Since Euler angles do not form such a space, the use of CRLBs to address orientation estimation is limited in its applicability. However, as long as the actual bounds are small relative to the range of the parameter space, the Cramér–Rao approach can be successfully used to produce reasonable results. For example, use of the CRLB should provide a sensible method of analysis for systems in which the calculated bound on an orientation angle is typically less than 10° . (In this paper, all calculated bounds are less than 1.0° .) However, when bounds exceed 10° , the accuracy of the results may become suspect. Of course, bounds greater than 360° are meaningless except for the implication that the measurement provides almost no information regarding the target's orientation! Related issues in which two or more orientations produce nearly indistinguishable images and are construed as equivalently the same target state have been discussed by Grenander *et al.*^{22,23} In this paper, it is assumed that *a priori* knowledge of the target's orientation state and the resolution of the sensor act to break any such degeneracies, so that they do not pose an issue.

When one is considering bounds on Euler angles, care must also be given to interpret them within the context of the relation between the actual and reference orientations of the target. This is due to the fact that the ordering of the Euler rotations used to specify an object's pose is not communicative. To make interpretation as simple as possible, we calculate the CRLBs presented in this paper by using the actual orientation of the object as the reference orientation. Given that the resulting bounds are small (in this case, less than 1.0°), differences between the ordering of rotations is negligible, and the bounds can be interpreted simply as the accuracy in yaw, pitch, and roll about the actual orientation.

3. TARGET/SENSOR MODEL

Section 2 demonstrated that a general statistical model for many imaging modalities and the corresponding CRLB for estimating a set of unknown parameters could be described by Eq. (14) and relation (15), respectively. Lurking within these simple formulas is the complicated function $g_q(\bar{x}|\xi)$, which describes the ensemble average of measurement $d_q(\bar{x})$ when the state of the target/sensor system is ξ . For a typical imaging system, $g_q(\bar{x}|\xi)$ must describe the 3-D relations between the sensed image and the target structure along with effects such as directional

lighting, viewing perspectives, radiometry, spectral effects, system point-spread function (PSF), properties of the detector elements (e.g., CCD FPA), and the bidirectional reflectivity distribution function (BRDF) describing the target's surface reflectance properties. Relation (15) also requires calculation of the derivative of the mean measurement at each FPA element as a function of each state parameter of interest, i.e., $\partial g_q(\bar{x}|\xi)/\partial \xi_k$. Self-occlusion and other lighting effects generally cause the projection of a 3-D structure onto a 2-D measurement to be highly nonlinear, making a closed-form solution intractable for all but the simplest of target structures. To solve more complicated problems, this paper examines the approach of making finite-difference approximations to the derivatives. This reduces the problem to that of constructing a numerical model for accurately rendering images of the target as would be seen by the sensor in the absence of noise. As will be discussed in detail below, this approach necessitated development of a novel rendering method in order to accurately calculate the required derivatives. Although this paper will focus on imaging sensors that measure the 2-D spatial intensity distribution of incoherent light reflected by a target (e.g., a typical camera), the approach taken could easily be extended to other imaging modalities such as thermal, hyperspectral, laser radar, and range-Doppler sensors.

A. Image-Rendering Algorithm

Two rendering approaches were investigated. The first was based on the established tracking for antisatellite (TASAT) code developed by the U.S. Air Force Research Laboratories.³⁴ As is common practice in many imaging codes, TASAT first generates a pristine image on a fine grid that is representative of the radiance reflected from the target toward the sensor. Convolution of the pristine image with the system PSF (also modeled on a fine grid) produces the intensity pattern incident on the FPA. The noise-free sensor measurement can then be obtained by downsampling.

As described in detail in Appendix A, by forcing each element of the pristine image grid to either be fully occupied by a segment of the target or be empty, this approach leads to a quantization of the position of edges of target substructures. Antialiasing measures in TASAT may be employed to reduce this effect—the result is a grid with a locally adaptive, *yet still finite*, cell spacing. The rendering approach used by TASAT is a proven method for generating radiometrically accurate images and possesses the advantages of being straightforward and of utilizing the efficiency of the fast Fourier transform for performing convolutions. However, as shown in Appendix A, the edge location quantization effect can cause large inaccuracies when approximating derivatives by using finite differences. In theory, this limitation can be overcome by using finer-resolution grids, but in practice the grid sizes become painfully large.

To circumvent this problem, we developed a novel target/sensor model based on populating the surfaces of the target with surface elements (as illustrated in Fig. 1) and by use of a fully analytic representation of the system PSF. Each surface element has associated data values, such as location, differential area, surface normal, and

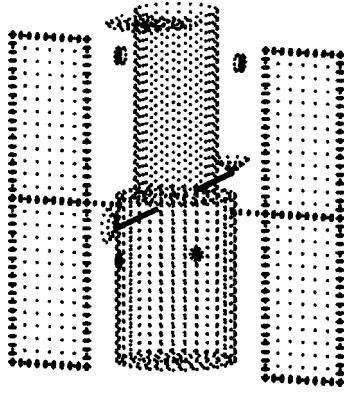


Fig. 1. 3-D plot of the centers of the surface elements used to populate a target's exterior. For purpose of illustration, the number of surface elements in the model shown here has been significantly reduced below that of a typical model.

Table 1. Sampling Density and Purpose of the Series of Grids Used by the Analytic Rendering Algorithm

Grid Type	Sample Spacing	Purpose
FPA grid G_{FPA}	FPA pixel spacing	Sensor measurement
Convolved image grid $G_{\text{conv img}}$	$\leq (\text{FPA pixel width})/4$ and $\leq (\text{PSF width})/4$	Convolved image (photon flux density incident on the detector)
Obscuration grid G_{obscur}	Finer sampling than $G_{\text{conv img}}$ and several samples across each target surface element	Determine self-obscurations of target segments, generates depth- and component-map by-products

material type. The various calculation grids employed by this analytic rendering algorithm are summarized in Table 1. The algorithmic procedure is performed through the following steps:

1. For the viewing perspective and for each directional light source, an obscuration mapping operation is performed in which the target surface elements are projected onto the obscuration grid G_{obscur} , facilitating classification of each element as being visible from that direction or as being obscured by other surface elements. Under this mapping, the surface elements are treated as having circular shapes with radii associated with their differential areas. The obscuration profiles are determined by the relation between the surface normal and the LOS. To determine obscurations of one surface element by another, G_{obscur} should be fine enough such that, when projected onto the target, the height and the width of each target surface element subtends two or more grid cells. As a result of this criterion and the criterion given in step 3 below, G_{obscur} will also be several times finer than that of the convolved image grid $G_{\text{conv img}}$ used in steps 3 and 4 below. By-products of this step are depth and component maps calculated on G_{obscur} . An example of a depth map is displayed in Fig. 2. The reader should understand that although partial obscuration of one surface element by an-

other is not considered, this step of the rendering process does not result in any type of quantization of the positions of the surface elements.

2. The radiant exitance of each illuminated and nonobscured surface element is calculated from its BRDF properties and the relative geometry associated with the normal to the surface and with the lighting and viewing directions.

3. With each target surface element treated as a point source, the optical intensity distribution incident on the FPA associated with an individual element α is calculated on the convolved image grid $G_{\text{conv img}}$. The optical intensity distribution is given by a closed-form expression describing the sensor's PSF, which is a function of the vector separation between the projection of the surface element α 's center onto $G_{\text{conv img}}$ and the center of the grid elements of $G_{\text{conv img}}$ themselves:

$$\begin{aligned} \text{image intensity distribution}(x', y')|_{(\text{surf. elem. } \alpha)} \\ = K \times (\text{radiant intensity})_{\alpha} \\ \times \text{psf}(x' - P_{x'}(\bar{z}_{\alpha}), y' - P_{y'}(\bar{z}_{\alpha})). \end{aligned} \quad (16)$$

In Eq. (16), $\bar{z}_{\alpha}$ denotes the position of surface element α in the target coordinates, $P_{x'}$ and $P_{y'}$ project the target coordinate frame onto the FPA plane, and the constant K includes all unit conversion and system throughput factors. The cell spacing of the convolved image grid $G_{\text{conv img}}$ used in this step is finer than that of the FPA and is such that the PSF is well sampled. This provides an accurate representation of the intensity distribution before the effects of sampling onto the FPA.

4. The results of the previous step are summed over all surface elements of the target to produce the photon flux distribution incident on the FPA. This data product is termed the convolved image.

5. The convolved image is downsampled onto the FPA pixel grid G_{FPA} , producing the noiseless FPA measurement. This grid is the same as that used in the equations of Subsection 2.A. An example is shown in Fig. 2.

Since both the projection operator and the PSF function in Eq. (16) are closed-form expressions, the influence of each surface element on the FPA measurement varies smoothly with changes in the element locations. Subsection 3.B demonstrates that this allows extremely accurate calculations of the required derivatives of the FPA measurements to be achieved with the use of finite-difference approximations.

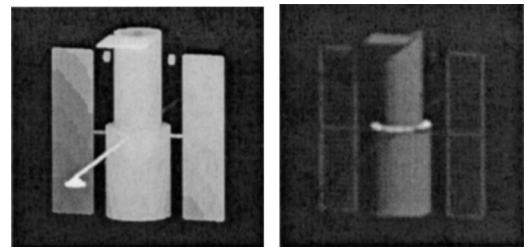


Fig. 2. Example of a depth map and corresponding radiometric rendering using the surface element target/sensor modeling approach.

B. Target Self-Obscuration

One difficulty remains unresolved by this model, which is an accurate treatment of the continuum of changes in the obscuration of one surface by another with respect to relative changes in viewing and lighting geometry. Our model does treat this effect to some degree by allowing a smooth and continuous variation of the location of the overlap between surfaces. For relatively simple objects superimposed on dark backgrounds, it may be reasonable to assume that the border of the silhouette is the predominant source of information regarding position and orientation. However, this issue undoubtedly warrants further exploration. One approach to assessing the significance of the possible error is to compare the Fisher information calculations obtained by using the analytic rendering code with a closed-form solution derived for a simple target consisting of two overlapping rectangular plates.³⁵ Section 4 will follow this approach in validating other aspects of the analytic rendering methodology described in this paper for performing CRLB analysis.

Some thought has been given to development of a scheme for modeling smooth and continuous obscuration of target features by each other. This goal might be accomplished by modifying the algorithm described above as follows³⁶:

1. The surface of each of the primitive shapes composing the target are tessellated into an interlocking set of triangles.

2. At a given orientation, each triangle is compared with all other triangles, areas of overlap are identified, and the unobscured area of each triangle is calculated. In performing this comparison, one uses a series of simple criteria to eliminate combinations of triangles that cannot overlap (such as separation transverse to and along the LOS). This reduces the need for more computationally burdensome region-of-overlap calculations to be performed for all (No. of elements)² combinations. Care must be given to address situations in which surface elements physically intersect and in which an element is obscured by multiple other elements.

3. As above, the radiance exitant from each surface element (i.e., in this case, each triangular facet) is treated as emanating from a point located at the center of the element.

4. A high density of surface elements samples the target such that the light distribution is effectively continuous relative to the PSF and IFOV widths.

4. ACCURACY VALIDATION OF THE TARGET/SENSOR MODEL

This section compares the accuracy of the numerical model against calculations from an exact closed-form solution for a simple target. These comparisons will be used to validate that the analytic rendering algorithm described in Section 3 performs as expected and to indicate the degree of accuracy that can be obtained in using it to compute CRLBs. For an arbitrary target, a closed-form expression of the mean FPA measurement is quite difficult to find. However, for a simple target such as a flat rectangular plate, the calculations are tractable. Two ro-

tation aspects are considered: in-plane orientation about the LOS and out-of-plane orientation around an axis perpendicular to the LOS. For both cases, we assume that the nominal orientation of the plate is such that it is perpendicular to the LOS and that the long edges of the rectangle are parallel to columns of the FPA array. The plate's surface is assumed to have a Lambertian BRDF and to be illuminated uniformly from all directions, i.e., ambientlike lighting conditions. The closed-form expression for the PSF is described by Eqs. (B4) and (B5) below, which result in a pyramidlike profile with square support. Note that since a function with finite support or discontinuous derivatives has infinite bandwidth, the sample spacing of the convolved image grid $G_{\text{conv img}}$ must approach zero to satisfy the Nyquist criterion. However, using a grid with cells $\leq 1/4$ the width of the PSF is found to produce image renderings that are not visibly aliased and that exhibit smooth and continuous changes in pixel values with respect to changes in target position and orientation.

A. Out-of-Plane Orientation

Under the described imaging geometry, the effect of an out-of-plane rotation is to cause a foreshortening of the projection of the plate onto the FPA. The angle of rotation is considered to be about the $\hat{y}$ axis, which corresponds to the vertical direction of the FPA and is perpendicular to the LOS. To simplify the math, we set the dimensions of the plate such that it is narrow along the $\hat{x}$ axis and extends greatly beyond the FPA's FOV along the $\hat{y}$ axis. As a result, the FPA image will not vary as a function of $\hat{y}$, reducing the derivation to a one-dimensional (1-D) calculation. Under the Lambertian BRDF assumption, the photon flux seen by the FPA from any portion of the plate remains constant as a function of orientation. The resulting source intensity distribution in the plane of the FPA and its derivatives are derived in Appendix B. Except for a proportionality constant, expressions (B9)–(B25) completely specify the Fisher information matrix of relation (15) in closed form.

We now compare the CRLB on estimating the plate's out-of-plane orientation as calculated numerically and by the exact closed-form expression. The imaging geometry is as described above, and details of the target/sensor system are given in Table 2.

Figure 3 shows the resulting FPA image and the derivative of the FPA image for the plate at $\theta_y = 20^\circ$, corresponding to rotation about the $\hat{y}$ axis by 20° from its nominal position of being perpendicular to the LOS. The dotted-dashed lines in Fig. 3(b) encircle the slice of elements used for the plots of Fig. 4. The values of the FPA measurements, their derivatives, and the single-pixel Fisher information

$$\frac{[\partial g(x, y | \theta_y) / \partial \theta_y]^2}{g(x, y | \theta_y) + \sigma_{x,y}^2} \quad (17)$$

are plotted in Figs. 4(a), 4(b), and 4(c), respectively, for the FPA elements along this horizontal slice. The total Fisher information is found by summing the term in relation (17) over all FPA elements. As derived in Subsection 2.A, the terms $g(x, y | \theta_y)$ and $\sigma_{x,y}^2$ correspond to the vari-

Table 2. Target/Sensor Parameters for Comparing the Numerical and Closed Expression Results for Imaging of a Flat Plate

System Parameter	Value
Aperture diameter	1.57 m
FPA IFOV	0.414 μrad
FPA size	32 $\times$ 32 pixels
PSF width	0.21, 0.41, 0.82, 1.7, 2.5 μrad
PSF profile	Triangle function
Target range	600 km
Target width W_R	3.5 m
Directional illumination	0 $\text{W m}^{-2} \lambda^{-1}$ (λ in μm)
Ambient lighting	500 $\text{W m}^{-2} \lambda^{-1}$ (λ in μm)
Exposure time	4 ms
Typical signal	24.7 $\times 10^6$ photoelectrons total, 49.4 $\times 10^3$ per/pixel on target
BRDF	Lambertian
Wavelength	0.65 μm
Bandwidth	0.2 μm
Analog-to-digital gain	1
Background noise	40e ⁻ rms
Throughput efficiency	0.05

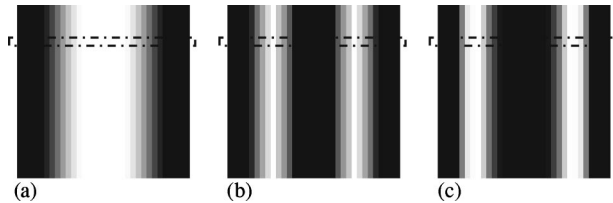


Fig. 3. (a) FPA image of the flat rectangular plate at a 20° out-of-plane rotation from the nominal position (face perpendicular to the LOS), (b) derivative of the image measurement with respect to out-of-plane rotational motion $\delta\theta_y$ about the vertical axis, (c) relative magnitude of the Fisher information at each pixel.

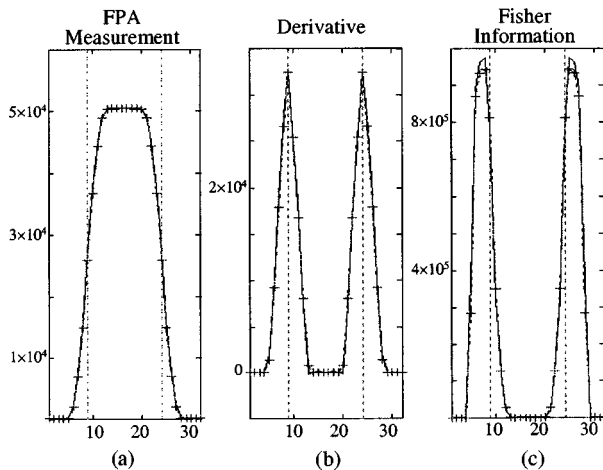


Fig. 4. Plots of the pixel values corresponding to the image slices indicated in Fig. 3. Calculations performed by using the exact closed-form expression and by using the numerical model are plotted with solid curves and a series of plus signs, respectively. Except for (c), the difference in the results is too small to be visible. The dashed vertical lines indicate the position of the edge of the target.

ances of the signal-dependent photon noise and background Gaussian noise sources (e.g., CCD-electronics-related read noise), respectively. As described in Table 2, the background Gaussian noise is $\sigma = 40e^-$, and $g(x)$, plotted in Fig. 3(a), is the average number of photoelectrons at each pixel (x, y) , e.g., typically on the order of 5×10^4 photoelectrons for pixels that lie on the target and corresponding to a photon noise level of approximately 224 photoelectrons.

The comparison between the numerical approximation and the closed-form expression results are quite good. Using the exact expression gives a CRLB equal to 0.0152°. The numerical approximation gives 0.0154°, indicating a discrepancy of 1.03%. Similar tests were done for a range of orientation angles, PSF widths, surface element densities, and sampling ratios between the grids $G_{\text{conv img}}$ and G_{FPA} used for the convolved and the FPA image. The results indicated that to obtain numerical results that are within 10% of the closed-form solutions, the following criteria on the various sampling rates associated with the target surface element density and the different calculation grids should be met:

1. The density of surface elements representing the target's surface should be such that the width of the main PSF lobe as projected onto the target subtends at least four surface elements.
2. The density of surface elements should satisfy a similar criterion for the projected height/width of a single FPA element, i.e., the IFOV height and width should each subtend at least four surface elements.
3. The convolved image grid $G_{\text{conv img}}$ should have four or more samples across the width of the PSF's main lobe and at least two samples across each FPA element.
4. The width of the PSF needs to be at least twice the width of a FPA element.

The first two criteria correspond directly to the intuitively obvious need for the target to be well sampled relative to the resolution of the image system in order to prevent aliasing effects. Similarly, the third criterion requires that the imaging model adequately sample the structure of the PSF. The last criterion is surprising and may indicate a limitation of the model. It may be that this last criterion can be relaxed if the first three are strengthened. Further investigation is warranted in understanding this issue.

For the 58 test cases that met these criteria, the mean accuracy of the CRLB calculation was 1.24% with a worst-case discrepancy of 6.4%. It was seen that improving the target and PSF sampling rates (i.e., by increasing the target surface element density and using a finer-grid spacing for the convolved image grid $G_{\text{conv img}}$) tended to improve the accuracy. Conversely, tests in which the above criteria were not quite met generally exhibited less agreement with the closed-form results. Over the full range of cases considered, in which the inequalities listed above were never broken by more than a factor of 4, accuracy decreased gracefully as the criteria were exceeded and aliasing-type effects increased. Further exploration is needed to understand and characterize the detailed relationships between the various sampling rates and the ac-

curacy of the numerical calculations. The main interest for this paper, though, is verification that sampling densities can be chosen for which the numerical CRLB calculations consistently agree with the exact closed-form expression within levels reasonable for a first-order analysis (e.g., within 10% or better).

It is somewhat surprising and of significant interest that the plot in Fig. 4(c) indicates that the single-pixel Fisher information peaks just beyond the outside edge of the plate (indicated by the dashed vertical lines). This is because for Poisson statistics the noise level is highest on the bright interior side of the target edge and drops toward the background noise level on the exterior side. As a result, expression (17) has a peak just beyond the edge of the target, at a point where the derivative is still high but the signal (and thus the noise) has decreased to a small value. Note that this phenomenon is directly related to the fact that the dominant noise for this case is signal dependent. If the measurement were of a low-reflectance target superimposed on a bright background and still photon noise dominated, the same phenomena would occur but with the Fisher information peak shifted to the interior side of the target edge. When the noise is predominantly signal independent and invariant of pixel location, so that the denominator of expression (17) is constant, the Fisher peak would lie, as expected, centered over the edge location. The width and the location of the Fisher peak are also influenced by the relative edge response of the system, which depends strongly on the ratio of PSF width to IFOV. It is apparent that the complex interplay among PSF, IFOV, background noise, and strength of the target signal has a significant effect on the Fisher information corresponding to an edge's location. Such phenomena will have a strong impact on the accuracy limits with which a system can be used to perform assessment tasks related to target feature positions. Using CRLBs to analyze and understand such influences on the accuracy limits for performing target assessment tasks provides a means for optimizing design and operation of sensors and associated estimation algorithms.

B. In-Plane Orientation

This subsection validates the numerical approach to CRLB calculation for estimating the in-plane orientation of a target (rotation about the LOS). Again, the validation is accomplished by comparison of the numerical results against exact closed-form expressions corresponding to a flat-plate target. These expressions are derived in Appendix B. The comparisons were repeated for a series of PSF widths and for several sampling density combinations in the numerical model. Typical results are displayed in Figs. 5 and 6. For this particular case, the exact CRLB was 0.0046°, and the numerical result was in error by 0.5%. Fifty-eight cases were tested that satisfied the criteria outlined in Subsection 4.A. The maximum discrepancy was 3%.

It is interesting to note that the CRLB for determining in-plane orientation is much lower than the out-of-plane CRLB bound calculated in Subsection 4.A. Comparing Figs. 3 and 5, we see that the derivatives of the FPA measurements near the plate's edge are higher for in-plane rotation than for out-of-plane rotation. As a result, the

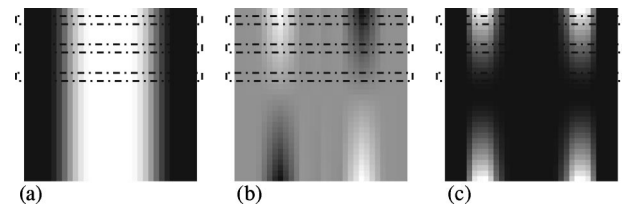


Fig. 5. (a) FPA image of the flat rectangular plate at its nominal position (face perpendicular to the LOS), (b) derivative of the image measurement with respect to in-plane rotational motion $\delta\theta_z$ about the LOS, (c) relative magnitude of the Fisher information at each pixel.

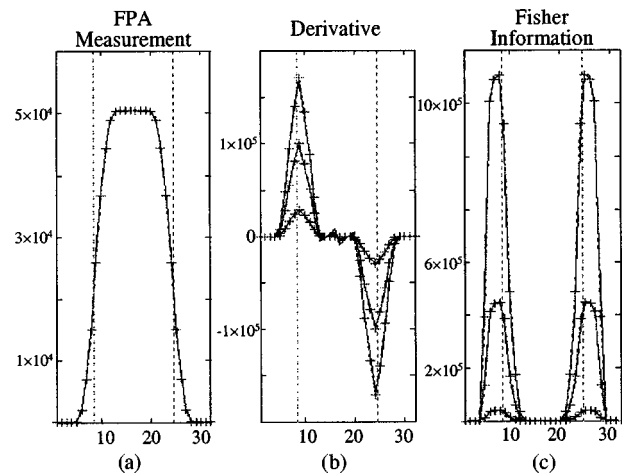


Fig. 6. Plots of the pixel values corresponding to the image slices indicated in Fig. 5. Calculations performed by using the exact closed-form expression and by using the numerical model are plotted with solid curves and a series of plus signs, respectively. As seen from the plots, the results agree almost perfectly.

Fisher information is stronger and the bound is lower. The magnitude of the information is also seen to increase with distance from the center of rotation, since the change in position induced by a differential change in attitude is proportional to this length. In Section 6, the trend for bounds on estimating in-plane orientation to be generally lower (allowing better accuracy) than those on estimating out-of-plane orientation is further demonstrated in an analysis example involving estimating the pose of the Hubble Space Telescope (HST).

The results just described correspond to a plate that is oriented such that its normal is close to parallel with the LOS. Similar comparisons were performed for a plate that was nearly on edge with respect to the LOS. The closed-form expressions are not presented here, but the derivation is analogous to that of Appendix B. Again, the numerical and exact results agreed well, with the largest error over the 52 cases considered being 8% and with a mean of approximately 4%.

C. Sensitivity of Difference Approximations to Perturbation Size

As discussed in Section 3 and Appendix A, there is potential for the calculations to be sensitive to the size of the perturbation used in the finite-difference approximations of the derivatives required by relation (15). If the perturbation to the system state is too large, the average FPA

measurement $g(x|\xi + \delta\xi)$ will not vary linearly in the perturbation $\delta\xi$. If it is too small, the numerical precision of the computer will become an issue. For the computations to be robust, it is desirable that there be a large range of magnitudes of $|\delta\xi|$ for which the derivative approximations are accurate and consistent.

We tested for these sensitivity issues by calculating the CRLB for 13 different target/sensor parameters over a large range of perturbation step sizes. The renderings were performed by using both the method described in Section 3 and a variant mode that consists of the following steps:

1. A high-resolution *pristine* rendering of the target is generated on a grid that has eight cells across each FPA element. This intermediate rendering does not yet include any blurring effects. Like the procedure described in Section 3, the target is modeled as a set of discrete surface elements. Lighting and obscuration calculations are made for the particular viewing geometry, and the results are multiplied by a system throughput factor to give the photon flux incident on the focal plane from each surface element. Each individual flux rate is added to the pristine grid at the cell corresponding to the projection of the surface element's center onto the focal plane. A final multiplicative factor adjusts for quantum efficiency and exposure time. Although the details differ, this rendering method is similar to the one used in TASAT in the sense that light reflected/emitted from facets of the target's surface is treated such that it emanates at the nearest point on a projected grid—i.e., the radiance is spatially quantized.

2. The system PSF distribution is calculated on a similar grid.

3. A *convolved* image corresponding to the photon flux distribution incident on the FPA array is produced by discrete convolution of the pristine image with the PSF.

4. Downsampling the convolved image and adding any background sources gives the noiseless FPA measurement.

We will refer to the mode just described as the grid-based rendering mode and the mode described in Section 3 as the analytic rendering mode.

A model of the HST is used for the target, and the sensor is modeled after the Maui Space Surveillance System's 1.6-m telescope and GEMINI imaging system. Details are given in Table 3. Note that to avoid exceeding the memory limits of the computer used for these computations, the modeled FPA elements are twice as wide as the elements of the true GEMINI system. However, since resolution was primarily limited by the blur size of the PSF, this discrepancy is probably minor.

Using both the analytic and grid-based rendering modes, we calculated CRLBs on the HST's orientation as a function of the size of the difference used in the derivative approximations. The results are plotted in Fig. 7. The analytic rendering approach is seen to be consistent for step sizes ranging from $\Delta\theta = 10^{-13}^\circ$ to 1.0° , which is 14 orders of magnitude! Selecting a finite-difference step size from the middle of this range gives a very good margin of comfort that the calculations will be accurate for a

large variety of imaging conditions. At the largest step size considered ($\Delta\theta = 10^\circ$), features in the image shift by several pixels. This is undoubtedly beyond the linear region, causing the magnitudes of the derivatives to be overestimated. As expected, the CRLB appears to err on the low side. For step sizes smaller than 10^{-13}° numerical errors become significant.

Results using the grid-based rendering method are in good agreement with the analytic approach for $\Delta\theta = 0.1^\circ$ to 1.0° , jump around a bit from 10^{-5}° to 0.01° , and exhibit a second region of stability from 10^{-12}° to 10^{-5}° . We suspect that for $\Delta\theta = 0.1^\circ$ to 1.0° , the step size is large enough that spatial quantization is not limiting the accuracy, but that from $\Delta\theta = 10^{-5}^\circ$ to 0.01° , the quantization effect is causing derivatives to be overestimated and as a result is artificially reducing the CRLB. At step

Table 3. Target/Sensor Parameters Used in Testing the Sensitivity of the CRLB Calculations to the Magnitude of the Perturbations Used in Approximating Derivatives by Finite Difference

System Parameter	Value
Aperture diameter	1.57 m
FPA IFOV	0.414 μrad
FPA size	64×64 pixels
PSF width	2.484 μrad
PSF profile	Hermite–Gaussian expansion
Target range	900 km
Target length	12.91 m
Solar illumination	$2 \times 10^3 \text{ W m}^{-2} \lambda^{-1}$ (λ in μm)
Ambient lighting (earthshine)	$500 \text{ W m}^{-2} \lambda^{-1}$ (λ in μm)
Exposure time	4 ms
Typical signal	5.0×10^7 photoelectrons total, 21×10^3 per pixel on target
BRDF	Lambertian
Wavelength	0.65 μm
Bandwidth	0.2 μm
Analog-to-digital gain	1
Background noise	$160e^-$ rms
Throughput efficiency	0.05

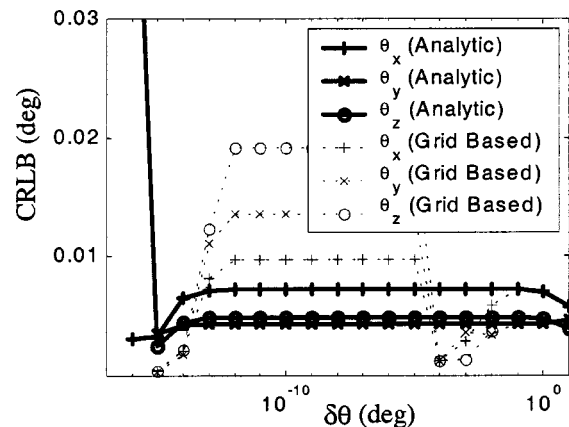


Fig. 7. Dependency of the target orientation CRLB calculation on the size of the difference used in the derivative approximations and on the method of rendering (analytic or grid based).

sizes of $\Delta\theta = 10^{-12}$ to 10^{-5} , the surface elements do not usually move enough to traverse any of the pristine grid boundaries. However, some effect must cause slight changes in the images; otherwise, the derivatives would all be zero, causing the calculated CRLB to be infinite. We suspect that the changes are due to BRDF-induced variations in reflected light. If this is the case, it indicates that this effect may contribute as strongly to the Fisher information as does feature location.

CRLBs were also calculated on estimating offsets to the FOV and on estimating the PSF width. The results are plotted in Figs. 8 and 9, respectively. Units of the ordinate axis are in fractions of a FPA element and in fractions of the nominal PSF width. For both bounds, the analytic rendering approach again gives consistent calculations over an enormous range of finite-difference step sizes. For the FPA offset, the grid-based results agree with the analytic approach at only $\Delta\theta = 0.01^\circ$ and 0.1° , and no other range of stability is exhibited. Selecting a value from this range would not leave much margin for error if the grid-based approach for image rendering were to be used for CRLB calculation. Although it never agrees with the analytic approach, the grid-based method produces consistent answers for the PSF width CRLB over a large range. This is because finite differences in the image due to changes in the PSF shape are not affected by locational quantization of target features. Discrepancies between the two curves in Fig. 9 are most likely attributed to the fact that the analytic rendering model calculations had been performed by using a triangularly shaped PSF profile while the grid-based renderings were produced by using a Gaussian PSF profile. Since the PSF profiles were different, moderate differences in the accuracy bound on estimating PSF width are to be expected.

A similar analysis is performed in Appendix A, using TASAT to produce the renderings required for the finite-difference calculations. Problems related to spatial quantization caused this approach to be even less robust to the choice of size than the grid-based rendering approach just described.

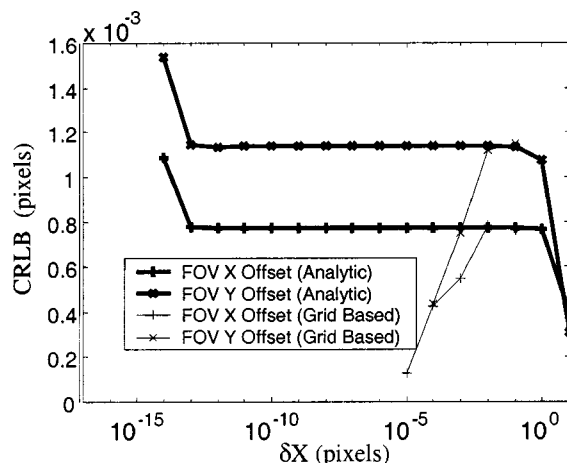


Fig. 8. Dependency of the FOV offset CRLB calculation on the size of the difference used in the derivative approximations and on the method of rendering.

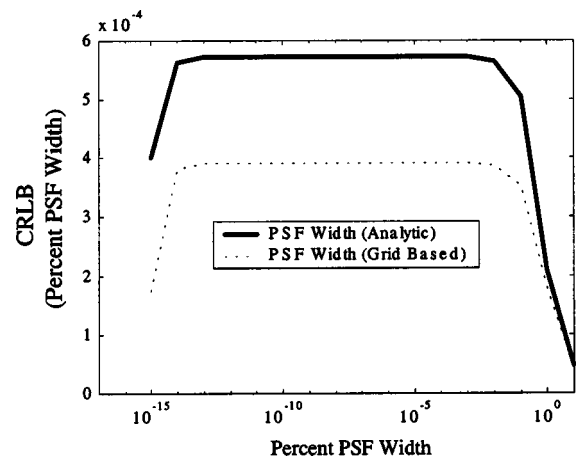


Fig. 9. Dependency of the PSF width CRLB calculation on the size of the difference used in the derivative approximations and on the method of rendering.

5. NUISANCE PARAMETERS

Typically, in addition to the primary target properties to be estimated, there are other unknown quantities that affect the measurement but are otherwise not of interest. Since these quantities serve only to complicate the problem, they are commonly referred to as nuisance parameters.¹⁹ The additional uncertainty related to the nuisance parameters tends to increase the CRLB for the quantities that are actually of interest. This occurs through cross correlations in the Fisher information. These correlations are expressed in the off-diagonal elements of the Fisher matrix and increase the CRLB when they differ from zero.¹⁹ As pointed out in Section 1, inclusion of all relevant nuisance parameters can be crucial to obtaining calculated bounds that are connected to real-world problems.

For assessment tasks such as target orientation, the list of possible nuisance parameters includes any unknown quantity that might affect the image. Examples of possible unknowns are details about the target's 3-D structure, reflective properties of its surface, and strength and direction of the illumination sources. For purpose of illustration, we assume that, other than orientation, the only quantities that are not well characterized are uncertainties in the PSF size and shape and in the relative position of the FOV with respect to the target's center (i.e., offsets to the LOS). We explored the effect of these nuisance parameters on target orientation for the imaging scenario detailed in Table 3 for a range of PSF widths and target orientations. Variations in the PSF shape were parameterized by using a sum of the first four terms of the 2-D Hermite-Gaussian sequence.³⁷ The impact of uncertainties in the PSF shape and size were surprisingly small. For all examples tested, their inclusion never caused the CRLB to increase by more than 20%. Variations in the offsets to the FOV had a stronger effect and in some cases increased the CRLB by up to a factor of 2. The influence of nuisance parameters related to uncertainties in target/sensor characteristics is explored in more detail by Gerwe and Idell.²⁷

It is also important to note that the orientation of the object is calculated in reference to the sensor's LOS. Al-

though in some cases the system's pointing angles relative to the earth and the orientation of the image about the LOS may be estimated from the image, more generally they will be provided by the mount control system. Any errors in these parameters will cause the estimate of the target's orientation to be biased. In this paper, the CRLBs on orientation estimation are always in reference to the true LOS.

6. MULTIPERSPECTIVE DATA FUSION

This section demonstrates the utility of the analytic rendering approach developed and validated in the previous sections for performing CRLB calculations. As an illustrative example, we analyze the influence of viewing perspective on estimation accuracy limits and show the potential for reducing the single-sensor CRLB by combining image data taken at multiple perspectives. The specific assessment task considered is the estimation of the HST's pose using three geographically separated sensing systems with characteristics similar to those of the 1.6-m telescope and the GEMINI sensor at the U.S. Air Force Research Laboratory's Maui Space Surveillance Site.

As portrayed in Fig. 10, the viewing geometry is such that the perspectives of sensors A, B, and C are along or

thogonal LOS vectors described by the $\hat{x}$, $\hat{y}$, and $\hat{z}$ axes, respectively. A second set of orthogonal body-centered axes is defined in reference to the HST, with $\hat{Y}$ directed

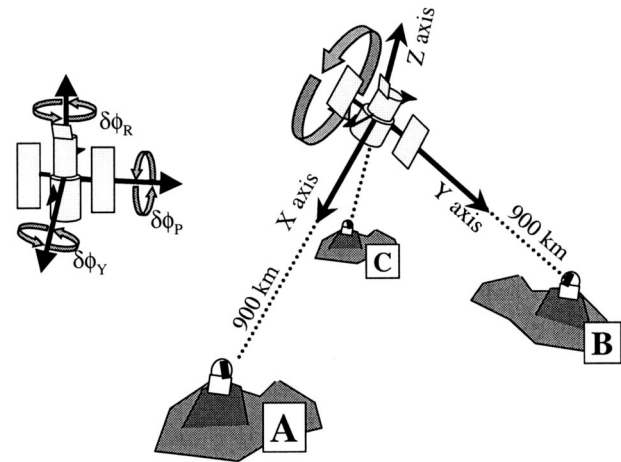


Fig. 10. Imaging geometry of sensors A, B, and C with respect to the target and diagram of the yaw, pitch, and roll body-centered orientation axes $\{\delta\phi_Y, \delta\phi_P, \delta\phi_R\}$. A series of orientations is considered as the HST is rotated around sensor B's LOS with its pitch axis coaligned with the $\hat{Y}$ axis.

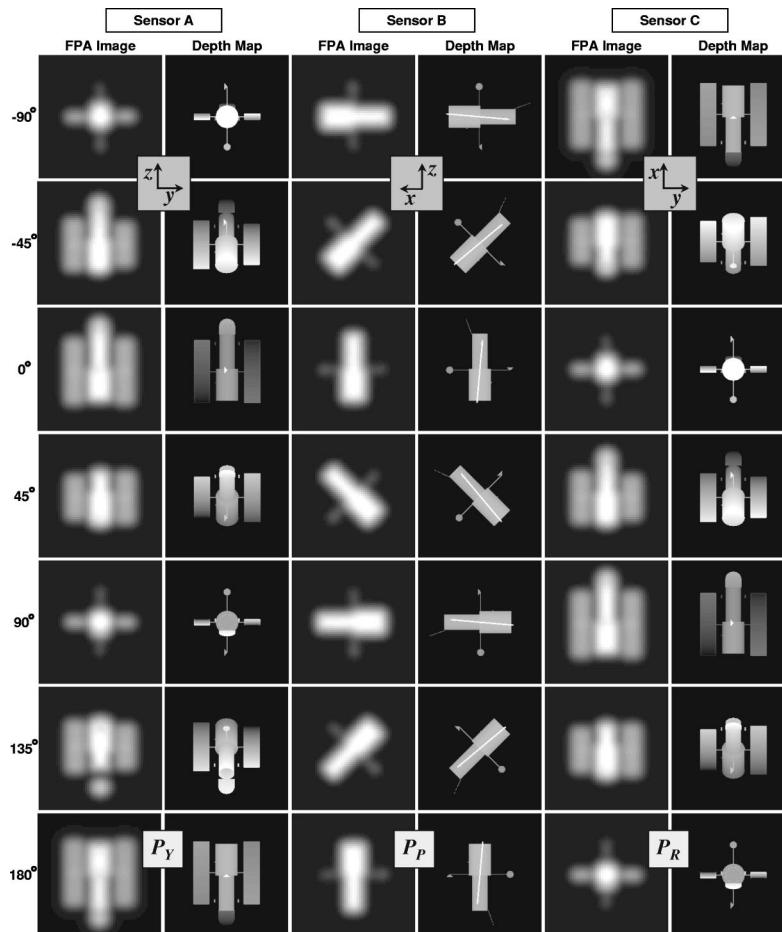


Fig. 11. Mean image measurements obtained by sensors A, B, and C at a series of orientations of the HST as it is rotated about sensor B's LOS or, equivalently, the HST's $\hat{P}$ axis (pitch). To assist interpretation of the imaging geometry, a corresponding set of depth maps is also displayed. The $\hat{x}$ -, $\hat{y}$ -, and $\hat{z}$ -axis directions (see Fig. 10) are overlaid for convenience. The three labels P_Y , P_P , and P_R denote particular viewing perspectives referred to later in Section 6.

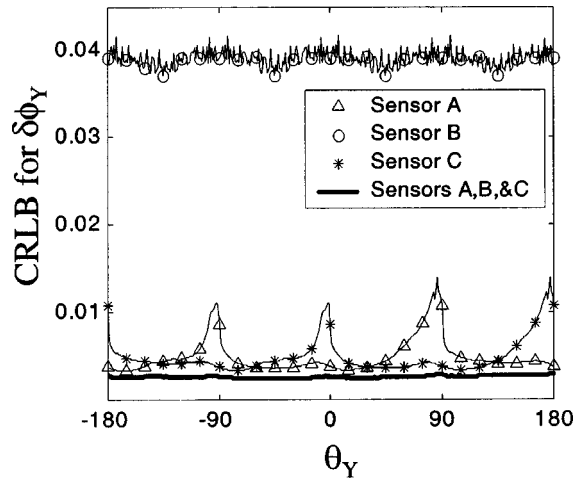


Fig. 12. CRLB on estimating the yaw aspect ($\delta\phi_Y$) of the HST's orientation with the use of image measurements from sensors A, B, and C, both individually and in combination. The bound is calculated for a sequence of orientations as the HST is rotated 360° about sensor B's LOS or, equivalently, the HST's $\hat{P}$ axis. As expected, the bounds vary as a function of the target/sensor geometry. Combining measurements from all three sensors always reduces the lower bound below that obtained for any single sensor.

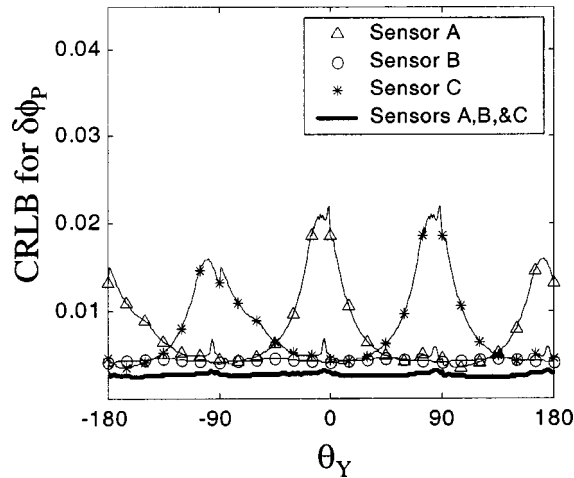


Fig. 13. CRLB on estimating the pitch aspect ($\delta\phi_P$) of the HST's orientation.

through the antennas, $\hat{P}$ directed along the struts connecting the solar panels to the satellite body, and $\hat{R}$ directed along the optic axis of its telescope. These will be referred to as the yaw, pitch, and roll axes, respectively.

The calculations are performed for a sequence of orientations (θ_y) as the HST is rotated about sensor B's LOS (i.e., the $\hat{y}$ axis or, equivalently, $\hat{P}$) (see Fig. 11). The reference pose ($\theta_y = 0^\circ$) is defined such that the HST has the optic axis of its telescope directed along the $\hat{z}$ direction and its antennas aligned with the $\hat{x}$ axis. This is equivalent to setting $\{\hat{Y} = \hat{x}, \hat{P} = \hat{y}, \hat{R} = \hat{z}\}$. CRLBs are calculated on rotational deviations in yaw, pitch, and roll, i.e., $\{\delta\phi_Y, \delta\phi_P, \delta\phi_R\}$, about the body-centered axis vectors. This is equivalent to using Euler angles but with the $\{0, 0, 0\}$ position always equal to the true orientation.

Calculating the CRLBs on angles referenced to another base orientation (e.g., on $\{\theta_x, \theta_y, \theta_z\}$) can make the results difficult to interpret because of nonuniqueness and of noncommunicability of the order in which rotations are applied.

The details for both sensors are identical to those described in Table 3. Vertical and horizontal offsets to the FOV of the FPA, and the height and the width of the PSF, are included as nuisance parameters. The results are plotted in Figs. 12–14. Immediate inspection reveals that, as expected, the bounds on the yaw, pitch, and roll aspects of the pose vary as a function of viewing perspective. At orientations near $\theta_y = -90^\circ$ and 90° , sensor C has the best leverage on $\delta\phi_Y$, and near $\theta_y = 0^\circ$ and 180° , sensor A has the upper hand. More extreme variations are seen in Figs. 13 and 14 for $\delta\phi_P$ and $\delta\phi_R$, respectively.

The bounds on orientation estimation with sensor B are roughly identical for the entire sequence of poses considered. This is because the viewing perspective is the same for all values of θ_y and as a result the images are almost identical within a rotation about the LOS. The only differences among them are slight details involving the relative spatial relations between target features and FPA element boundaries. However, the small fluctuations in the calculated values are probably just as likely to be attributable to accuracy limitations of the target/sensor model as to the influence of the location of the FPA boundaries on the CRLB.

The plots also clearly demonstrate that the CRLB for the combined measurement is always superior to that of any sensor individually.²⁷ The potential benefit is obvious. Multiple perspectives reduce the occurrence of “blind spots” at which the combined measurement is insensitive to an aspect of the satellite's pose.

Further examination of the results provides some insight as to which perspectives and target features provide the best orientation-related information. Let us start with $\delta\phi_Y$. As labeled in Fig. 11, we will refer to the perspectives of sensors A, B, and C at $\theta_y = 180^\circ$ as the P_Y , P_P , and P_R perspectives, respectively. We also differentiate between two regimes of foreshortening effects that occur when out-of-plane rotation alters the appearance of a broad surface as projected onto the sensor's FPA. Both are instances of the same effect, but since the projected area of a surface is roughly $A \cos(\hat{n}_{\text{surf}} \cdot \hat{n}_{\text{LOS}})$, the rate at which the projected area changes depends on the angle between the LOS $\hat{n}_{\text{LOS}}$ and the normal to the surface, $\hat{n}_{\text{surf}}$. When $\hat{n}_{\text{surf}}$ points in the general direction of the sensor, out-of-plane rotation will cause a weak expansion or contraction of the projected area. This projection geometry will be referred to as a foreshortening of type I. When $\hat{n}_{\text{surf}}$ is generally perpendicular to the LOS, out-of-plane rotation will cause a similar but much stronger effect on the surface's silhouette. This regime of foreshortening will be called type II. In either case, the magnitude of the change in appearance will be proportional to the width of the surface along the direction perpendicular to both $\hat{n}_{\text{surf}}$ and the axis of the rotational aspect under consideration.

In the P_Y perspective, changes in yaw correspond to in-plane rotations of the satellite and result in the best (lowest) CRLB, $\sigma_{\delta\phi_Y} = 0.0038^\circ$. Figure 15 displays a map of

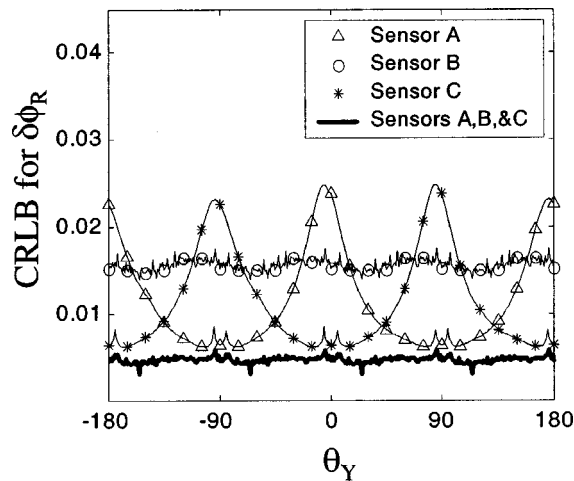


Fig. 14. CRLB on estimating the roll aspect ($\delta\phi_R$) of the HST's orientation.

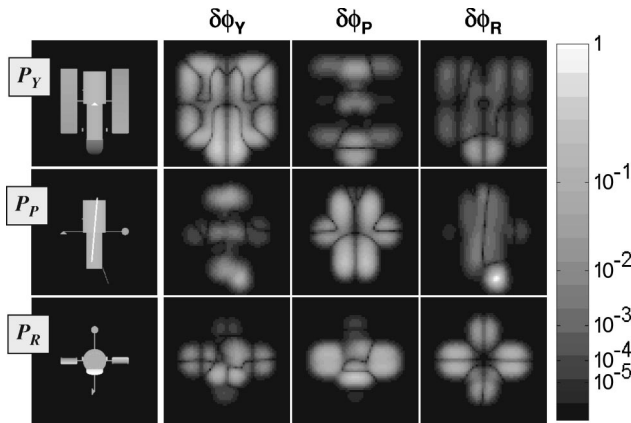


Fig. 15. These images portray the relative strength of the Fisher information provided by each pixel of a FPA regarding the yaw, pitch, and roll orientation aspects of the HST. The top, middle, and bottom rows correspond to the viewing perspectives seen by sensors A, B, and C, respectively, with the HST positioned in its $\theta = 180^\circ$ pose. These perspectives are referred to as P_Y , P_P , and P_R and are identical to those seen in the bottom row of Fig. 11.

the relative strength of the Fisher information gained from each FPA element. Examination indicates that for this perspective, the in-plane rotation of the edges of the solar panels and the silhouette of the satellite body are very informative about yaw. Estimation of $\delta\phi_Y$ is more difficult from perspective P_R , which results in $\sigma_{\delta\phi_Y} = 0.0108^\circ$. In this case, most of the Fisher information is gained from a combination of translational movement and type II foreshortening of the solar panels, of the sides of the main body, and of the telescope shutter. From perspective P_P , weak yaw information is gained from type I foreshortening of the silhouette of the body corresponding to out-of-plane rotational orientation. A combination of changes in the shutter's projected appearance also contributes to the Fisher information. This perspective is relatively uninformative compared with P_Y and P_R , and it results in the worst CRLB, $\sigma_{\delta\phi_Y} = 0.0392^\circ$.

P_P and P_R give roughly equal leverage for estimating

pitch, with CRLBs of $\sigma_{\delta\phi_P} = 0.0042^\circ$ and 0.0046° , respectively. Examining Fig. 15, we see that for the P_P perspective, in-plane orientation aspects of the silhouette of the satellite body provide strong information about pitch. The strongest contributions in the P_R perspective are type II foreshortening of the solar panels and of the telescope shutter. Differences in the tilt of each solar panel and of the shutter cause the CRLB to actually be slightly lower at an orientation approximately 5° off P_R . Perspective P_Y is the least informative for estimating pitch, with a CRLB of $\sigma_{\delta\phi_P} = 0.0133^\circ$. In this case, the primary hints at pitch are type I foreshortening of the solar panels and of the sides of the body, both of which are relatively weak.

As also seen with $\delta\phi_Y$ and $\delta\phi_P$, the optimum orientation for estimating $\delta\phi_R$ occurs when this orientational aspect corresponds to in-plane rotation of the target's pose, i.e., when the rotational axis is parallel with the LOS. It appears that the Fisher information regarding in-plane orientation tends to be strong relative to that of the other orientational aspects and as such should be easier to estimate. Of course, this trend will be highly subject to other factors such as target shape and lighting properties. In this particular case, the features that contribute the most information are the projected position of the solar panels and of the antenna on the FPA array. The second-best perspective for estimating roll is P_P . Here the Fisher information is strongly dominated by type II foreshortening of the shutter. Without this feature, the CRLB would be much higher (worse accuracy), since the satellite body and solar panels provide almost no clues to the roll aspect. Least favored is perspective P_Y , for which the main information source is the weaker type I foreshortening observed on the solar panels and on the telescope shutter.

Most of the insights demonstrated by this illustrative example agree with what our intuitive feel would predict to be the best perspectives for determining different orientational aspects of the target. However, the perspectives discussed were chosen to be along lines of symmetry of the target, which tends to simplify interpretation of the results. More realistic examples would include additional complexities such as directional lighting effects, higher level of detail in the target, and more complicated geometries.

One obvious utility of this type of analysis is to help plan the optimal viewing geometry for estimating parameters of interest. As a hypothetical example, one might be able to predict that the most advantageous perspective for estimating the cross-track orientation of a low-earth-orbit satellite with a ground-based telescope would be achieved shortly after it rises past the horizon. Furthermore, optimal lighting might be found to occur when the sun is as close as possible to being directly behind the sensor. This would indicate that the images are best taken at morning terminator conditions (with the target in the west). Cross-track orientation might turn out to be most easily estimated at the satellite's culmination. If it could be assumed that the satellite was not maneuvering or if a prior was used to describe constraints on the satellite's maneuvering capabilities, data fusion benefits gained from joint estimation of all orientation aspects and from

the use of a full sequence of pictures taken throughout the pass would probably greatly reduce the CRLB.

We must comment that the calculated orientation accuracy bounds seem small compared with the resolution of the simulated imagery. However, implicit in the calculations of these bounds is the assumption that the 3-D structure and material properties of the target/sensor characteristics are known *exactly*. As described in Section 5, the only nuisance parameters included in the simulations of this paper are target position and PSF size and shape. As shown by Gerwe and Idell,²⁷ including a more realistic and comprehensive set of target- and sensor-related uncertainties may significantly increase the CRLB.

7. SUMMARY

Fundamental limits on the accuracy with which target properties can be estimated from imaging methods were explored through an approach based on the CRLB. The paper focused on orientation estimation using conventional imaging of incoherent light sources. However, the methodology is equally applicable to other assessment tasks and sensing modalities. A novel rendering technique was developed based on representation of the 3-D structure of a target's surface by populating it with a large number of surface elements. This representation circumvented difficulties that were encountered in approximating derivatives with finite differences. By enabling a continuum in the relations between the location of one surface's edge with respect to another, varying the target's orientation caused smooth and continuous changes in the rendered image. Differential changes with which one surface obscures another were not, however, accounted for. Further exploration is needed to determine whether this effect is small enough that it is reasonable to ignore its contribution to the total Fisher information.³⁶

The modeling approach was validated by comparing the numerical results against numbers obtained from an exact closed-form expression derived for a simple rectangular plate. The results indicated that the CRLB calculations for orientation estimation could be expected to be accurate to within 6%. We also explored the degree with which inclusion of a few nuisance parameters increased the CRLB. Uncertainties in the structure of the PSF were found to have a surprisingly small effect, but, under some conditions, uncertainties regarding the target's location relative to the FOV increased the CRLB by a factor of 2.

An illustrative example was presented, demonstrating the utility of this approach for understanding the influence of various system parameters on estimation accuracy limits. Specifically, we explored how the informational content of an image measurement regarding a target's 3-D orientation could be optimized by choice of viewing perspective. Maps indicating the relative strength of Fisher information as a function of location in the image provided insights in understanding how location and orientation of different target features relative to viewing perspective influenced their contribution to the total information. The potential benefit that could be de-

rived through the combined information gained from images taken at multiple viewpoint perspectives was also shown.

APPENDIX A: EFFECT OF EDGE LOCATION QUANTIFICATION ON FINITE-DIFFERENCE APPROXIMATIONS

The aim of this appendix is to demonstrate the difficulties encountered when attempting to make finite-difference derivative approximations based on rendering codes that have the effect of quantizing the position of the edges of target substructures. In this context, the term edge is used to denote either the true edge of a flat facet or the apparent edge of a curved surface such as a sphere, which appears in its projection onto a 2-D plane.

TASAT is a good example of a prominently used code with the quantization properties just described. TASAT is commonly used to simulate tracking and imaging of satellites and other high-altitude bodies. TASAT was used in the first attempt at performing the CRLB calculations described in this paper (until the associated problems in accurately calculating derivatives was discovered). This code models the target as a construct of abstract geometric primitives (rectangle, cylinder, sphere, cone, etc.), each with a set of assigned material properties. Rendering is effectively performed through the following steps³⁴:

1. Calculate a pristine radiance map on a fine grid that is representative of the radiance emitted from the target toward the sensor. This is performed by

- (a) tracing rays toward the target through the corners of each grid cell,
- (b) determining the location at which each ray first touches one of the target's primitives and noting the material type of the intersected surface,
- (c) associating radiance values with each ray according to the BRDF of the materials and the light source direction(s),
- (d) obtaining values at the center of each cell by averaging the radiances from the rays at each corner, and,
- (e) as an optional antialiasing measure, if the values at the corners of a cell differ by more than 10%–30%, throwing additional rays from within that cell in order to more accurately sample the radiance variations.

2. Perform a discrete convolution of the pristine image with a PSF corresponding to the effects of blurring by the optics, jitter, etc. The result corresponds to the intensity pattern incident on the FPA.

3. Downsample from the fine grid to a grid matching the spacing of the FPA detector elements.

As briefly pointed out in Section 3, the antialiasing steps 1(d) and 1(e) above help to reduce aliasing effects by a local adaptation of the grid resolution to the level of target detail and by a small degree of low-pass filtering provided through averaging of rays at the corners of each grid cell to get the value in the center. However, despite this measure, spatial location is still quantized to finite step sizes,

and it is precisely this fact that causes difficulties in finite-difference derivative approximations.

Consider the following example. Derivatives of the noiseless FPA measurements with respect to the orientation angle θ_z are approximated by (1) rendering the target at slightly perturbed orientations about the nominal position, (2) taking the difference, and (3) dividing by the magnitude of the perturbation:

$$\frac{\partial g(x|\theta_z)}{\partial \theta_z} \approx \frac{g(x|\theta_z + \delta\theta_z/2) - g(x|\theta_z - \delta\theta_z/2)}{\delta\theta_z}. \quad (\text{A1})$$

As portrayed in Fig. 16, let us suppose that the gross effect of the slight rotation $\delta\theta_z$ on the upper right edge of the plate is for it to shift to the right by approximately 1 pixel as projected onto the FPA. Assume a system PSF with a top-hat-like shape and a width of approximately 4

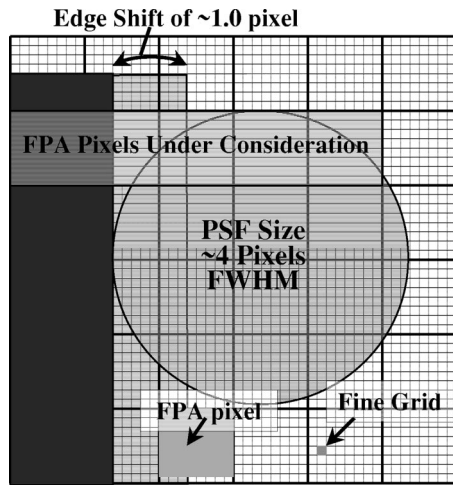


Fig. 16. Pristine image projection of the upper right corner of a flat plate as it is rotated by $\delta\theta$, causing the edges to shift approximately 1 FPA pixel to the right. The fine lines correspond to the pristine grid, with the FPA element edges highlighted by the thicker grid lines. The illustration also contains an overlay of the PSF spot and a highlighted row of FPA elements.

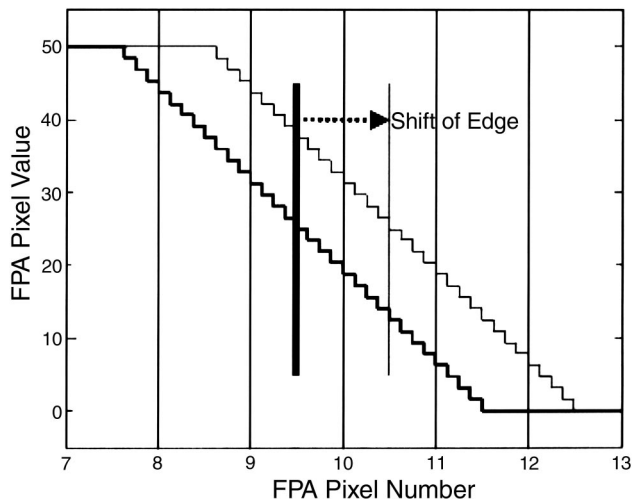


Fig. 17. Change in the simulated intensity distribution along a row of FPA elements as the edge of a target structure in Fig. 16 shifts approximately 1 FPA element to the right. The numbers on the ordinate axis denote FPA pixel edges, and the small steps indicate edges of the convolved image grid elements.

FPA pixels. Let us see how rotations of the plate change the simulated FPA measurements when the fine-resolution grid used for the pristine and convolved images has eight subcells across each FPA element. A plot of the expected response along the horizontal line of FPA elements highlighted in Fig. 16 is shown in Fig. 17.

As the plate is rotated smoothly from $\theta_z - \delta\theta_z/2$ to $\theta_z + \delta\theta_z/2$, the edge shifts to the right through approximately eight fine-grid elements. The values of the elements of the simulated convolved image near the moving edge undergo a corresponding eight step changes. However, depending on the exact amount of the total rotation $\delta\theta_z$ and the relative position of the edge on the grid, the number of step changes could range from 7 to 9. The accuracy of the finite-difference approximation will be only 1 part in 8, or approximately 12.5%. By virtue of causing edges to shift across more pixels, better accuracy can be obtained by choosing larger values of $\delta\theta_z$. However, too large a value will also start to introduce effects from higher-order derivative terms and will invalidate the finite-difference approximation.

A much worse case can occur for target edges near the center of rotation at which spatial shifts will be very small. Most edges will not shift enough to cross any fine-grid boundaries and thus will have zero contribution to the derivative approximations. This is unlikely to be a severe problem, since the actual effect of these edges on the CRLB should be small anyway. However, depending on the exact value of $\delta\theta_z$ and the relative position of the edge to the grid lines, there will be occasional occurrences for which an edge moves by a minuscule amount yet crosses a grid boundary. The resulting effect will be a grossly overestimated magnitude in the finite-difference approximation. It is clearly evident that an exceedingly high grid resolution will be needed to comfortably ensure that a value of $\delta\theta_z$ can be chosen that produces adequate accuracy for the calculations at *all* regions of the FPA and for all target features.

With the use of TASAT, the CRLB on estimating the orientation of the HST about its LOS was calculated by using a range of finite-difference step sizes. The results provide a clear demonstration of the difficulties in accurately approximating derivatives when using the approach employed by TASAT to produce renderings. (Similar calculations were performed in Subsection 3.C to demonstrate the robustness of the analytic rendering method to the choice of finite-difference step size.) The renderings were performed with the antialiasing features in TASAT enabled. The intermediate pristine and convolved images were calculated on grids with eight points across each IFOV. The signal strength was higher, but otherwise the sensor characteristics and the target geometry were similar to those outlined in Table 3. Nuisance parameters were not included, and only the in-plane orientation aspect was considered. The results are plotted in Fig. 18. As the size of the difference used in the derivative approximations decreases below 0.1° , the quantization effect quickly causes severe overestimation of the derivatives. Above 1° , we expect higher-order derivatives to begin to significantly affect the approximations. This leaves a slim region between 0.1° and 1° , where one might hope that the derivatives have been accurately cal-

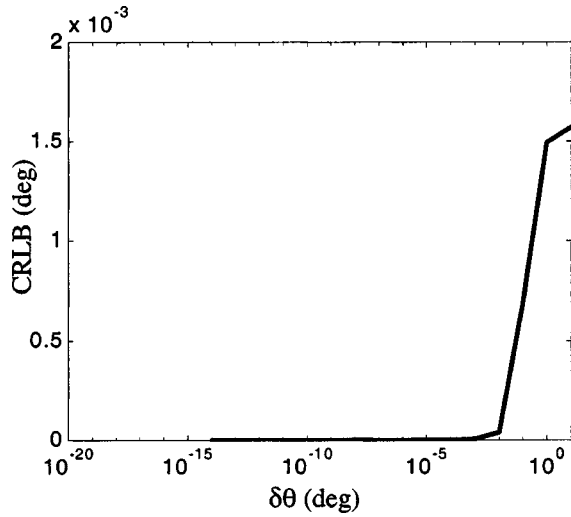


Fig. 18. Dependency of the target orientation CRLB calculation on the size of the difference used in the derivative approximations when using TASAT to render the images.

culated both near the center of rotation and near the edges of the FOV. Of course, this range is expected to vary for different imaging geometries and sensor characteristics. Obviously, ensuring that the calculations are accurate will be difficult and will require re-evaluation of an appropriate step size range each time the target, sensor, or imaging geometry characteristic is significantly altered.

APPENDIX B: CLOSED-FORM EXPRESSIONS FOR A FOCAL-PLANE-ARRAY IMAGE OF A FLAT-PLATE TARGET

This appendix derives closed-form expressions for the flux incident on the elements of a FPA from a flat-plate target and its derivative as a function of the plate's orientation. Two rotational aspects are considered: in-plane orientation about the LOS and out-of-plane orientation around an axis perpendicular to the LOS. For both cases, we assume that the nominal orientation of the plate is such that it is perpendicular to the LOS and that the long edges of the rectangle are parallel to columns of the FPA array. The plate's surface is treated as having a Lambertian BRDF and as being illuminated uniformly from all directions, i.e., ambientlike lighting conditions. Equations (B4) and (B5) below provide a closed-form expression for the PSF corresponding to a pyramidlike profile with square support.

1. Out-of-Plane Orientation

Under the described imaging geometry, the effect of an out-of-plane rotation is to cause a foreshortening of the projected width of the plate onto the FPA. The angle of rotation is considered to be about the y axis. To simplify the math, we set the dimensions of the plate such that it is narrow along the $\hat{x}$ axis and extends greatly beyond the FPA's FOV along the $\hat{y}$ axis. As a result, the FPA image will not vary as a function of $\hat{y}$, reducing the derivation to a 1-D calculation. Under the Lambertian BRDF assumption, the radiant intensity (corresponding to incident photon

tons per steradian) seen by the FPA from any portion of the plate remains constant as a function of orientation. The resulting source intensity distribution in the plane of the FPA can now be expressed by the equations that follow:

$$\text{plate intensity}(x, y|\theta_y) \propto \text{rect}\left[\frac{x}{W_o(\theta_y)}\right], \quad (\text{B1})$$

$$W_o(\theta_y) = W_R \cos \theta_y, \quad (\text{B2})$$

$$\text{rect}(x) = \begin{cases} 0, & |x| > 1/2 \\ 1, & |x| \leq 1/2 \end{cases} \quad (\text{B3})$$

The quantity $W_o(\theta_y)$ is the projected width of the plate as seen on the FPA when rotated out of plane by θ_y . W_R is the true width of the plate.

The PSF is represented by a 2-D triangle function of width W_p in x and in y :

$$\text{psf}(x, y) = \text{tri}\left(\frac{x}{W_p}\right) \text{tri}\left(\frac{y}{W_p}\right), \quad (\text{B4})$$

$$\text{tri}(x) = \begin{cases} 1 + x, & -1 \leq x \leq 0 \\ 1 - x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad (\text{B5})$$

The integrated photon flux g at each FPA element (x, y) is the convolution of the projected source distribution and the PSF, integrated over the element area. If we denote $\mathcal{A}_{x,y}$ as the set of points within the region subtended by FPA element (x, y) , then

$$g(x, y) \propto \iint_{(\xi, \eta) \in \mathcal{A}_{x,y}} d\xi d\eta \iint d\xi' d\eta' \times \text{psf}(\xi - \xi', \eta - \eta') \text{plate intensity}(\xi', \eta'). \quad (\text{B6})$$

If we set the unit distance equal to the detector element spacing, $(\xi, \eta) \in \mathcal{A}_{x,y}$ is true when $\text{rect}(x - \xi) = 1$ (assuming square FPA elements and a 100% fill factor). Multiplying the integrand by this rect function allows us to change the limits of the integral over ξ and η to $\pm\infty$. Substituting relation (B1) and Eq. (B4) into relation (B6), we obtain

$$g_o(x, y) \propto \left\{ \int d\xi \text{rect}(x - \xi) \int d\xi' \times \text{tri}\left(\frac{\xi - \xi'}{W_p}\right) \text{rect}\left[\frac{\xi'}{W_o(\theta_y)}\right] \right\} \times \left\{ \int d\eta \text{rect}(y - \eta) \int d\eta' \text{tri}\left(\frac{\eta - \eta'}{W_p}\right) \right\}. \quad (\text{B7})$$

The subscript o in g_o has been added to denote that the quantity corresponds to the expected FPA photon fluxes for out-of-plane rotations of the plate. The second bracketed term is not dependent on the source orientation and evaluates to a constant, giving

$$g_o(x, y) \propto \int d\xi \text{rect}(x - \xi) \int d\xi' \times \text{tri}\left(\frac{\xi - \xi'}{W_p}\right) \text{rect}\left[\frac{\xi'}{W_o(\theta_y)}\right]. \quad (\text{B8})$$

A closed-form expression for the mean measurement at FPA element (x, y) is now found by substituting Eqs. (B2), (B3), and (B5) into relation (B8)—a straightforward but tedious exercise. The form of the solution depends on the magnitudes of W_p and $W_o(\theta_y)$. For $W_p > 1$ and $W_o(\theta_y) > 2W_p + 1$, the following result is obtained:

$$g_o(x, y) \propto \begin{cases} 0, & W_p + 0.5 \leq \chi \\ \mathcal{I}1(\chi), & W_p - 0.5 \leq \chi \leq W_p + 0.5 \\ \mathcal{I}2(\chi), & 0.5 \leq \chi \leq W_p - 0.5 \\ \mathcal{I}3(\chi), & -0.5 \leq \chi \leq 0.5 \\ \mathcal{I}4(\chi), & -W_p + 0.5 \leq \chi \leq -0.5 \\ \mathcal{I}5(\chi), & -W_p - 0.5 \leq \chi \leq -W_p + 0.5 \\ \mathcal{I}6(\chi), & \chi \leq -W_p - 0.5 \\ \chi = \chi_0 \equiv |x| - W_o(\theta_y)/2 \end{cases}, \quad (\text{B9})$$

The functions $\mathcal{I}1$ – $\mathcal{I}6$ are defined as follows:

$$\mathcal{I}1(\chi) = T1(W_p + 0.5) - T1(\chi), \quad (\text{B10})$$

$$\mathcal{I}2(\chi) = T1(W_p + 0.5) - T1(W_p - 0.5) + T2(W_p - 0.5) - T2(\chi), \quad (\text{B11})$$

$$\mathcal{I}3(\chi) = T1(W_p + 0.5) - T1(W_p - 0.5) + T2(W_p - 0.5) - T2(0.5) + T3(0.5) - T3(\chi), \quad (\text{B12})$$

$$\mathcal{I}4(\chi) = T1(W_p + 0.5) - T1(W_p - 0.5) + T2(W_p - 0.5) - T2(0.5) + T3(0.5) - T3(-0.5) + T4(-0.5) - T4(\chi), \quad (\text{B13})$$

$$\mathcal{I}5(\chi) = T1(W_p + 0.5) - T1(W_p - 0.5) + T2(W_p - 0.5) - T2(0.5) + T3(0.5) - T3(-0.5) + T4(-0.5) - T4(-W_p + 0.5) + T5(-W_p + 0.5) - T5(\chi), \quad (\text{B14})$$

$$\mathcal{I}6(\chi) = T1(W_p + 0.5) - T1(W_p - 0.5) + T2(W_p - 0.5) - T2(0.5) + T3(0.5) - T3(-0.5) + T4(-0.5) - T4(-W_p + 0.5) + T5(-W_p + 0.5) - T5(-W_p - 0.5), \quad (\text{B15})$$

where

$$T1(\zeta) = \frac{(W_p + 0.5)^2 \zeta - (W_p + 0.5) \zeta^2 + \zeta^3/3}{2W_p^2}, \quad (\text{B16})$$

$$T2(\zeta) = \frac{W_p \zeta - 0.5 \zeta^2}{W_p^2}, \quad (\text{B17})$$

$$T3(\zeta) = \frac{\zeta}{W_p} - \frac{\zeta}{4W_p^2} - \frac{\zeta^3}{3W_p^2}, \quad (\text{B18})$$

$$T4(\zeta) = \frac{W_p \zeta + 0.5 \zeta^2}{W_p^2}, \quad (\text{B19})$$

$$T5(\zeta) = \frac{(W_p + 0.5)^2 \zeta + (W_p + 0.5) \zeta^2 + \zeta^3/3}{2W_p^2}. \quad (\text{B20})$$

Taking the derivative of relation (B9) with respect to $W_o(\theta_y)$ gives

$$\frac{\partial g_o(x, y)}{\partial W_o} \propto \begin{cases} 0, & W_p + 0.5 \leq \chi \\ \mathcal{D}1(\chi), & W_p - 0.5 \leq \chi \leq W_p + 0.5 \\ \mathcal{D}2(\chi), & 0.5 \leq \chi \leq W_p - 0.5 \\ \mathcal{D}3(\chi), & -0.5 \leq \chi \leq 0.5 \\ \mathcal{D}2(\chi), & -W_p + 0.5 \leq \chi \leq -0.5 \\ \mathcal{D}1(\chi), & -W_p - 0.5 \leq \chi \leq -W_p + 0.5 \\ 0, & \chi \leq -W_p - 0.5 \\ \chi = \chi_0 \equiv |x| - W_o(\theta_y)/2 \end{cases}, \quad (\text{B21})$$

with functions $\mathcal{D}1$ – $\mathcal{D}3$ defined as

$$\mathcal{D}1(\chi) = \frac{(W_p + 0.5 - \chi)^2}{4W_p^2}, \quad (\text{B22})$$

$$\mathcal{D}2(\chi) = \frac{W_p - \chi}{2W_p^2}, \quad (\text{B23})$$

$$\mathcal{D}3(\chi) = \frac{2W_p - 0.5 - 2\chi^2}{4W_p^2}. \quad (\text{B24})$$

The derivative with respect to θ_y is easily found by applying the chain rule to Eq. (B2) and relation (B21):

$$\begin{aligned} \frac{\partial g_o(x, y|\theta_y)}{\partial \theta_y} &= \frac{\partial g_o(x, y|\theta_y)}{\partial W_o(\theta_y)} \frac{\partial W_o(\theta_y)}{\partial \theta_y} \\ &= -\frac{\partial g_o(x, y|\theta_y)}{\partial W_o(\theta_y)} W_R \sin(\theta_y). \end{aligned} \quad (\text{B25})$$

Except for a proportionality constant, expressions (B9)–(B25) completely specify the Fisher information matrix of relation (15) in closed form.

2. In-Plane Orientation

This subsection will derive an exact expression for the image as a function of the in-plane orientation of the plate about the LOS. As a function of orientation, the projected profile can be found by transforming the equations for the lines that form the left and right edges of the plate. In the nominal ($\theta_y = 0^\circ$) orientation, these equations are $x_o = \pm W_R/2$. The in-plane rotational transformation to orientation angle θ_z is

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} \cos \theta_z & -\sin \theta_z \\ \sin \theta_z & \cos \theta_z \end{bmatrix} \begin{pmatrix} x_o \\ y_o \end{pmatrix}. \quad (\text{B26})$$

Substituting $x_o = \pm W_R/2$ and solving for x as a function of y produces

$$x = \frac{\pm W_R/2 - y \sin \theta_z}{\cos \theta_z} \quad (\text{B27})$$

As projected onto the FPA, the resulting intensity distribution function for the plate is

$$\text{plate intensity}(x, y|\theta_z) = \text{rect} \left[\frac{|x|}{W_i(x, y|\theta_z)} \right] \quad \text{for } \theta_z: H \sin \theta_z < W_R, \quad (\text{B28})$$

$$W_i(x, y|\theta_z) = \frac{W_R - \text{sign}(x)2y \sin \theta_z}{\cos \theta_z} \quad (\text{B29})$$

$$\text{sign}(x) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases}, \quad (\text{B30})$$

where H is the height of the FPA and the units are in FPA elements. Substituting Eqs. (B4) and (B28) into relation (B6) gives the mean expected FPA measurement:

$$\begin{aligned} g_i(x, y|\theta_z) &\propto \int d\xi \text{rect}(x - \xi) \int d\xi' \left\{ \text{tri} \left(\frac{\xi - \xi'}{W_p} \right) \right. \\ &\quad \times \int d\eta \text{rect}(y - \eta) \int d\eta' \text{tri} \left(\frac{\eta - \eta'}{W_p} \right) \\ &\quad \left. \times \text{rect} \left[\frac{|\xi'|}{W_i(x, y|\theta_z)} \right] \right\}, \quad (\text{B31}) \end{aligned}$$

where we have used the subscript i on g_i to denote that this quantity corresponds to in-plane rotations of the plate.

Assuming that the orientation angle θ_z is near zero and that $W_R \gg 1$, then the plate's intensity distribution is slowly varying in η' and it is reasonable to take this term, $\text{rect}[|\xi'|/W_i(x, y|\theta_z)]$, outside the two most interior integrals. Within this approximation, the result is analogous to relation (B7), obtained for the case of out-of-plane rotations, and the integrals in η and η' evaluate to a constant, giving

$$\begin{aligned} g_i(x, y|\theta_z) &\propto \int d\xi \text{rect}(x - \xi) \int d\xi' \\ &\quad \times \text{tri} \left(\frac{\xi - \xi'}{W_p} \right) \text{rect} \left[\frac{|\xi'|}{W_i(x, y|\theta_z)} \right]. \quad (\text{B32}) \end{aligned}$$

By substituting Eqs. (B3) and (B5) into relation (B32), we obtain the final closed-form expression for the expected FPA flux at each element (x, y) for in-plane rotations and its derivative with respect to $W_i(x, y|\theta_z)$, i.e., $\partial g_i(x, y|\theta_z)/\partial W_i(x, y|\theta_z)$. For $W_p > 1$ and $W_R - H \sin \theta_z > 2W_p + 1$, these expressions are identical to relations (B9) and (B21), except that the quantity χ must be redefined to correspond to edge positions of the plate for in-plane rotations:

$$\xi = \chi_i \equiv |x| - W_i(x, y, \theta_z)/2. \quad (\text{B33})$$

The functions $I1$ – $I6$ and $D1$ – $D3$ are defined just as above.

Applying the chain rule, we find the derivative of g_i with respect to θ_z to be

$$\begin{aligned} \frac{\partial g_i(x, y|\theta_z)}{\partial \theta_z} &= \frac{\partial g_i(x, y|\theta_z)}{\partial W_i(x, y|\theta_z)} \frac{\partial W_i(x, y|\theta_z)}{\partial \theta_z} \\ &= \frac{\partial g_i(x, y|\theta_z)}{\partial W_i(x, y|\theta_z)} \times \left\{ -\text{sign}(x)y \right. \\ &\quad \left. + \frac{[W_R/2 - \text{sign}(x)y \cos \theta_z] \sin \theta_z}{\cos^2 \theta_z} \right\}. \quad (\text{B34}) \end{aligned}$$

The Fisher information at each pixel of the imaged plate can now be specified for in-plane rotations within a multiplicative constant by substituting the new definition of χ expressed by Eq. (B33) into expression (B9), (B21), and (B34).

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29. The bound given in relation (4) corresponds to the lowest MMSE achievable by an optimal estimator for the specific value of ξ used in the calculations. Another, more global bound can be calculated by averaging Eq. (3) over all values of ξ and weighted by the *a priori* distribution $p(\xi)$. This Bayesian bound gives the minimum-mean-square accuracy achievable by any estimator including those that are biased. See pp. 72–73 and 84–85 of Van Trees.¹⁸
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35. Such a comparison was performed after the compilation of this paper. The details of the calculations are too long to include here but were similar to those presented in Section 4 and Appendix B. It was found that the numerical CRLB calculations were generally 5%–20% larger than that indicated by the closed-form expression. This difference is too small relative to the potential inaccuracies in the numerical calculations to infer much about how the obscuration effects influence the Fisher information. It does, however, provide an indication that the overall effect is fairly small and that the relative location of target edges dominates the Fisher information.
36. The described modifications to the rendering algorithm were implemented subsequent to the compilation of this paper. Tests indicate that as a complex 3-D target was rotated, the pixel values in the vicinity of obscuration edges changed smoothly and continuously. A rigorous evaluation of the accuracy of the approach for computing CRLBs has not been performed, but preliminary results are promising.
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